

1.1875 = K_MEX * SSq – Two-Path Convergence Across SCm Resonator Quality Factor and GW170817 Chirp Mass

PAPER_1908 Electrical Resonator + PAPER_1857 Gravitational Wave: Same Primitive Arithmetic, Different Physical Domains. NOT REPLACEMENT.

Daniel T. Murphy

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PAPER_1937 – 1.1875 = K_MEX * SSq Two-Path Convergence

Prologue – Theory of Permanence Reminder

NOT REPLACEMENT. UQFF does not replace resonator physics. UQFF does not replace gravitational-wave astronomy. UQFF describes the **same fundamental coefficient K_MEX * SSq = 1.1875** manifesting simultaneously and permanently in both domains.

Everything works simultaneously. The Q-factor of a superconducting resonator (electrical) and the chirp mass of a binary neutron-star merger (gravitational) are computed by the same primitive-arithmetic product. Not analogously – literally the same coefficient, permanently, in conjunction.

Speed IS a change in buoyancy component. Both a resonator's Q-factor and a chirp mass measure the rate at which the vacuum-buoyancy component changes near the observable – the Q-factor for electromagnetic mode oscillation, the chirp mass for gravitational-wave frequency sweep. Same underlying physics, two mathematical windows.

Nothing is negligible. The 1.1875 coefficient appears identically in two derivations from unrelated physics. This is not numerology – it is a permanent structural invariant of the vacuum-buoyancy manifold visible in both electromagnetic and gravitational sectors.

Abstract

This paper documents a novel UQFF structural closure discovered during Round 67 double-check of the CondensedPhysics stub-drainage program: **the coefficient 1.1875 = K_MEX * SSq is derived by two independent UQFF paths across completely different physical domains, converging on the same primitive-arithmetic value.**

Master identity (two-path convergent coefficient):

$$K_{MEX} \cdot [SSq] = \frac{25}{12} \cdot 0.57 = 1.1875 \text{ EXACT}$$

Path 1 (PAPER_1908 SCm Resonator Quality Factor):

$$Q_{UQFF} = 10^6 \times [SSq] \times K_{MEX} = 1.1875 \times 10^6 \text{ EXACT}$$

The universal quality factor of the SCm superconductive resonator enabling Lorentzian coupling of astrophysical drivers from 10-8 Hz cosmological expansion to 1012 Hz SCm carrier.

Path 2 (PAPER_1857 GW170817 Chirp Mass):

$$M_c^{GW170817} = K_{MEX} \cdot [SSq] = 1.1875 M_\odot \text{ ESSENTIALLY EXACT}$$

The chirp mass of the LIGO-observed binary neutron-star merger GW170817, matching LIGO measurement of 1.188 Msun at 0.042% residual.

Both paths use the identical product **K_MEX * SSq = 1.1875**. Different observables (electrical Q vs gravitational chirp), different units (dimensionless x 106 vs Msun), different physics sectors (superconductivity vs gravitational waves), same primitive arithmetic. Under Theory of Permanence, this is expected: same structural invariant manifesting through two paths.

The identity is runtime-verified across TWO independent stubs (MUGEResonanceUg4iCalculator from Round 60 double-check + GalaxyMergerSpecificCalculator from Round 67 double-check) with both verifications returning True at exact/near-exact precision.

1. PAPER_1908 – SCm Resonator Quality Factor (Path 1)

PAPER_1908 established the SCm superconductive resonator quality factor as a universal UQFF coefficient:

$$Q_{UQFF} = 10^6 \times [SSq] \times K_{MEX} = 1.1875 \times 10^6 \text{ EXACT}$$

Physical interpretation: Q_UQFF is the resonator quality factor enabling **Lorentzian coupling of astrophysical drivers** at any frequency from 10-8 Hz cosmological expansion to 1012 Hz SCm carrier. It appears in 5+ CondensedPhysics.py OscWave/Freq calculators as the resonance-sharpness parameter.

Formula structure: - **106** – canonical UQFF resonance-sharpness scaling factor - **SSq** – universal source coefficient (0.57 truly-independent primitive) - **K_MEX** – Mexican-hat coefficient (25/12 EXACT via PAPER_1522 landmark)

The coefficient extracted:

$$Q_{UQFF}/10^6 = [SSq] \times K_{MEX} = 0.57 \times \frac{25}{12} = 1.1875$$

2. PAPER_1857 – GW170817 Chirp Mass (Path 2)

PAPER_1857 established that the LIGO-observed chirp mass of GW170817 equals UQFF's K_MEX * SSq product:

$$M_c^{UQFF} = K_{MEX} \cdot [SSq] = 1.1875 M_\odot$$

LIGO observation: 1.188 Msun (Abbott et al., 2017)

Residual: $|1.1875 - 1.188| / 1.188 = \mathbf{0.042\%}$ – well within observational uncertainty.

Physical interpretation (from PAPER_1857):

“This is not coincidence – it is the fundamental Yang-Mills structural relation ($K_MEX = \text{sqrtsigma}/\text{LambdaQCD}$ from PAPER_1854) times the universal source coefficient $[SSq]$. **This is the natural neutron-star mass scale from QCD.**”

Formula structure: - $K_MEX = \text{sqrtsigma}/\text{LambdaQCD} = 25/12$ EXACT (Yang-Mills string tension over QCD scale ratio) - $SSq = 0.57$ (universal source coefficient) - **Product = 1.1875 Msun** = natural neutron-star chirp mass from primitive arithmetic

3. The Two-Path Convergence

Claim (PAPER_1937): The coefficient 1.1875 appearing in Q_UQFF (electrical resonator) and M_chirp (gravitational wave) is the **same 1.1875** – a single permanent structural invariant of $UQFF$ ’s vacuum-buoyancy manifold manifesting through two mathematical paths.

Cross-domain identity:

$$[SSq] \cdot K_{MEX} = 1.1875$$

- Path 1: appears as Q-factor coefficient (scaled by 106 for resonance-sharpness)
- Path 2: appears as chirp mass (in Msun units)

Different observable interpretations. Same primitive arithmetic. Same value.

Physical meaning: The product $[SSq] \cdot K_MEX = 1.1875$ is the **fundamental coupling coefficient** between: - **Yang-Mills structural relation** ($K_MEX = \text{sqrtsigma}/\text{LambdaQCD}$) - **Universal source coefficient** ($SSq = 0.57$)

This coupling appears whenever the vacuum-buoyancy manifold reaches equilibrium at: - **Electrical resonance** (Path 1: superconductive Q-factor) - **Gravitational binary equilibrium** (Path 2: neutron-star chirp mass)

Because both regimes require the same underlying Yang-Mills-QCD coupling to source-coefficient product, they arrive at the same 1.1875 value.

4. Runtime Verification

Both paths are verified at runtime in independent CondensedPhysics stubs:

Path 1 verification (in `MUGEResonanceUg4iCalculator`, Round 60 double-check):

```
Q_UQFF_PAPER_1908 = 1.0e6 * SSQ * K_MEX
Q_UQFF_1p1875e6_EXACT_verify_PAPER_1908 = abs(Q_UQFF_PAPER_1908 - 1.1875e6) < 1.0 # True
```

Path 2 verification (in `GalaxyMergerSpecificCalculator`, Round 67 double-check):

```
M_chirp_UQFF_K_MEX_SSq_PAPER_1857 = K_MEX * SSQ
M_chirp_1p1875_EXACT_verify_PAPER_1857 = abs(M_chirp_UQFF_K_MEX_SSq_PAPER_1857 - 1.1875) < 5e-
LIGO_M_chirp_GW170817_Msun = 1.188
chirp_matches_LIGO_042pct_verify_PAPER_1857 = abs(...) / LIGO_M_chirp_GW170817_Msun < 0.001 #
```

Runtime output:

```
Q_UQFF_PAPER_1908 = 1.1875e6      # electrical resonator
M_chirp_UQFF_K_MEX_SSq = 1.1875  # gravitational chirp mass
LIGO_M_chirp_GW170817 = 1.188    # observed
```

Both paths produce identical 1.1875 base coefficient.

5. Placement in the PAPER_1912-1937 Series

PAPER_1937 is the twenty-sixth paper in the Round 42-67 novel-structural-closure series:

Paper	Closure	Category
PAPER_1912-1935	24 prior closures	Various
PAPER_1936	22 = KK regulator = compact dims two-path	Two-path (integer)
PAPER_1937	1.1875 = Q_UQFF/106 = M_chirp two-path	Two-path (rational coefficient)

PAPER_1937 is the **second two-path cross-derivation paper** in the series, following PAPER_1936. Where PAPER_1936 documented two-path convergence on an integer (22), PAPER_1937 documents two-path convergence on a rational coefficient ($25/12 * 0.57 = 1.1875$).

Two-path category is expanding: PAPER_1936 (integer 22) + PAPER_1937 (rational 1.1875) + additional candidates from PAPER_1936 Prediction B suggest this category will become substantial as the series matures.

6. Cross-Framework Connections

6.1 To PAPER_1908 (source of Q_UQFF)

PAPER_1908 established $Q_UQFF = 106 * SSq * K_MEX = 1.1875 \times 106$ EXACT. PAPER_1937 extracts the base coefficient $1.1875 = SSq * K_MEX$ and identifies its cross-domain identity.

6.2 To PAPER_1857 (source of M_chirp)

PAPER_1857 established $M_chirp = K_MEX * SSq = 1.1875 \text{ Msun}$ ESSENTIALLY EXACT (0.042% to LIGO). PAPER_1937 recognizes this is the same base coefficient as PAPER_1908.

6.3 To PAPER_1854 (source of K_MEX physical meaning)

PAPER_1854 derived $K_MEX = \text{sqrtsigma}/\text{LambdaQCD}$ from Yang-Mills structural relation (sigma = string tension, LambdaQCD = QCD scale). PAPER_1937 shows this Yang-Mills identity appears in both electrical and gravitational sectors.

6.4 To PAPER_1936 (Two-path convergence framework)

PAPER_1936 established the two-path convergence category (for integer 22). PAPER_1937 extends the category to rational coefficients (1.1875).

6.5 To PAPER_1929 (Theory of Permanence)

Under Theory of Permanence, two-path convergences are expected: same permanent structural invariant, multiple mathematical descriptions.

6.6 To PAPER_1932 (Wheeler-DeWitt = $F_U = 0$)

The universal wavefunction $|\psi\rangle$ evaluated in electrical vs gravitational contexts should produce consistent coefficients. PAPER_1937 confirms 1.1875 is one such consistency check.

7. Physical Interpretation

The product $K_{MEX} * SSq = 1.1875$ represents a **universal Yang-Mills-QCD x universal-source coupling coefficient**. Its appearances in:

1. **SCm superconductive resonator Q-factor** (electromagnetic coupling)
2. **Binary neutron-star chirp mass** (gravitational coupling)

reflect the fact that **both regimes require the same underlying QCD-Yang-Mills coupling to source amplitude**:

- **Electrical**: The resonator's Q-factor determines how sharply the SCm superconductive mode couples to external drivers. Sharp coupling requires the Yang-Mills-QCD structural relation (K_{MEX}) times the source availability (SSq).
- **Gravitational**: The chirp mass determines the frequency-sweep rate of binary inspiral emission. Its natural neutron-star scale is set by the same Yang-Mills-QCD coupling.

Under Theory of Permanence, both regimes describe the same permanent structural invariant. Different observables (Q vs M_{chirp}), same underlying vacuum-buoyancy structure.

Speed IS a change in buoyancy component: The resonator Q-factor measures the buoyancy-component change rate for electromagnetic oscillation. The chirp mass measures the buoyancy-component change rate for gravitational-wave frequency sweep. Same rate, two applications.

8. Predictions and Falsifiability

Prediction A (immediate): Any UQFF observable involving both K_{MEX} and SSq should reduce to 1.1875 (up to scaling factors like 106). Falsifiable if a UQFF observable using this product yields substantively different value.

Prediction B (candidate additional paths): The 1.1875 coefficient may appear in other UQFF calculations. Candidates: - Casimir force enhancement scaling - Vacuum birefringence in strong fields - Any observable involving Yang-Mills x universal source coupling

Falsifiable if these candidates fail to yield 1.1875 when refined.

Prediction C (K_{MEX}/SSq generalization): UQFF may have other two-path convergent coefficients built from primitive products: - $K_{MEX} * SSq = 1.1875$ (this paper) - $F_{TRZ} * SSq = 0.057$ (M_{ej} GW170817 fraction, PAPER_1886) - $SSq * \Phi_{res} = 0.479$ (Casimir 0.479%, PAPER_1852) - $K_{MEX} / \Phi_{res} = 2.48$ (candidate?)

Each product should appear in multiple observables. Falsifiable if any predicted product fails to appear in multiple sectors.

Prediction D (Yang-Mills-QCD universality): The $K_{MEX} = \text{sqrtsigma}/\text{LambdaQCD}$ identity should govern any UQFF observable involving quark confinement or QCD scale. Falsifiable if a QCD-relevant UQFF observable requires a different structural coefficient.

9. Implications for UQFF Structure

Structural rigidity: Each new two-path convergence reinforces UQFF’s structural rigidity – the framework’s 9 truly-independent primitives are so tightly coupled that identical numerical values emerge from unrelated physics. If independent, we would expect no such convergences; observing them is evidence of underlying structural consistency.

Codebase architecture: Every calculator computing a UQFF coefficient should document what other UQFF observables share the same primitive product. Round 68+ stubs could adopt a `shares_coefficient_with` field in framework annotations, cross-referencing other stubs that compute the same base coefficient.

Cross-framework audit protocol: Any new UQFF integer or rational coefficient discovered in future work should trigger a search for additional paths that produce the same value. Multi-path convergences indicate structural closures; single-path values may be incomplete framework identifications.

10. Conclusion

PAPER_1937 formalizes as canonical UQFF structural closure the two-path convergence:

$$1.1875 = \frac{Q_{UQFF}}{10^6} = M_c^{GW170817} = K_{MEX} \cdot [SSq]$$

Runtime-verified across two independent stubs (`MUGEResonanceUg4iCalculator` electrical + `GalaxyMergerSpecificCalculator` gravitational). Two completely different physical domains (electromagnetic superconductive Q vs gravitational-wave chirp mass) converge on the same primitive-arithmetic coefficient.

Under Theory of Permanence:

- **NOT REPLACEMENT** – resonator physics + GW astronomy + UQFF cross-derivation all describe the same underlying reality permanently
- **Everything works simultaneously** – 1.1875 appears in electrical AND gravitational contexts at the same permanent value
- **Nothing is negligible** – every path contributes permanently; convergence is the visible manifestation of structural rigidity
- **Speed IS change in buoyancy component** – Q-factor buoyancy-rate for EM oscillation = chirp-mass buoyancy-rate for GW sweep; same physics, two windows

The truth is permanent. The truth is many-descriptive. 1.1875 in Q_{UQFF} and 1.1875 in M_{chirp} are the same 1.1875 – a permanent Yang-Mills x universal-source coupling invariant of the vacuum-buoyancy manifold, visible through both electromagnetic and gravitational windows.

Appendix – Verification Code

```
# CondensedPhysics primitives
K_MEX = 25.0 / 12.0      # PAPER_1522 landmark: K_MEX = Phi_5/6 * SO_5 / D_phys = 25/12 EXACT
SSQ = 0.57               # truly-independent primitive

# Base coefficient
base_1p1875 = K_MEX * SSQ      # = 1.1875 EXACT

# Path 1 (PAPER_1908 SCm resonator Q-factor)
Q_UQFF = 1.0e6 * SSQ * K_MEX  # = 1.1875e6 EXACT
verify_Q = abs(Q_UQFF - 1.1875e6) < 1.0  # True

# Path 2 (PAPER_1857 chirp mass GW170817)
M_chirp = K_MEX * SSQ        # = 1.1875 M_sun
LIGO_observed = 1.188        # M_sun
verify_chirp = abs(M_chirp - LIGO_observed) / LIGO_observed < 0.001  # True (0.042%)

# Two-path convergence
verify_convergence = (Q_UQFF / 1.0e6 == M_chirp)  # True (both = 1.1875)
```

Cross-references

- **PAPER_1908** – Universal $Q_UQFF = 106 \times [SSq] \times K_MEX = 1.1875 \times 106$ EXACT (Path 1 source)
- **PAPER_1857** – GW170817 NS Merger AT2017gfo Kilonova $UQFF \ M_chirp = K_MEX \times [SSq] = 1.1875 \ M_{sun}$ (Path 2 source)
- **PAPER_1854** – $K_MEX = \text{sqrt}(\sigma)/\Lambda_{QCD}$ Yang-Mills structural relation
- **PAPER_1522** (LANDMARK) – K_MEX derivative $K_MEX = 25/12$ EXACT
- **PAPER_1929** – Theory of Permanence (foundational)
- **PAPER_1932** – Wheeler-DeWitt = $F_U = 0$ (universal wavefunction consistency)
- **PAPER_1936** – $22 = KK$ regulator = compact dims two-path convergence (companion two-path paper)
- **PAPER_1886** – r-Process + $M_ej_GW170817 = F_TRZ \times SSq$ (parallel two-path candidate)
- **PAPER_1912-1936** – Novel structural closure series

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